

seeed studio

INDUSTRIAL PRODUCT DATASHEET

[2%off] reBot Arm B601-DM Bundle with Jetson Thor

Fully assembled, open-source 6+1 DoF robotic arm bundle with NVIDIA® Jetson AGX Thor™ for edge VLA deployment, visual grasping, teleoperation, and precision automation.

SKU

E26052001

UPDATED

2026-09-11 13:06:23



Overview



The reBot Arm B601-DM Bundle with Jetson Thor is a fully assembled, 100% open-source 6+1 DoF robotic arm paired with the NVIDIA® Jetson AGX Thor™ Developer Kit for on-site VLA model deployment such as GR00T, PI, and OpenVLA. It supports ROS1/2, Isaac Sim, and LeRobot, and is suited for applications including voice control, GraspNet visual grasping, teleoperation, assistive automation, and precision tasks in harsh or constrained environments.

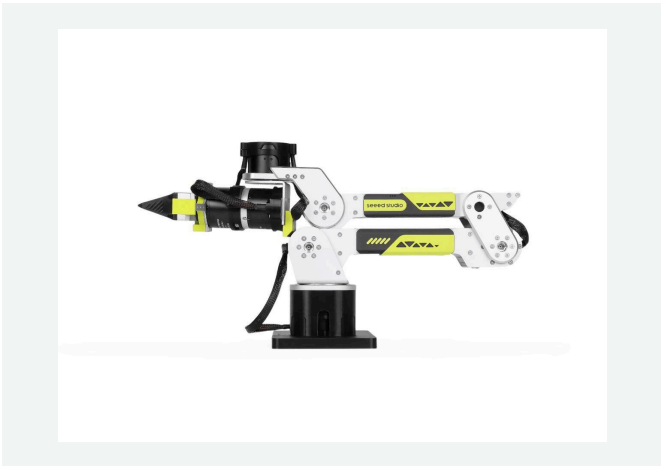
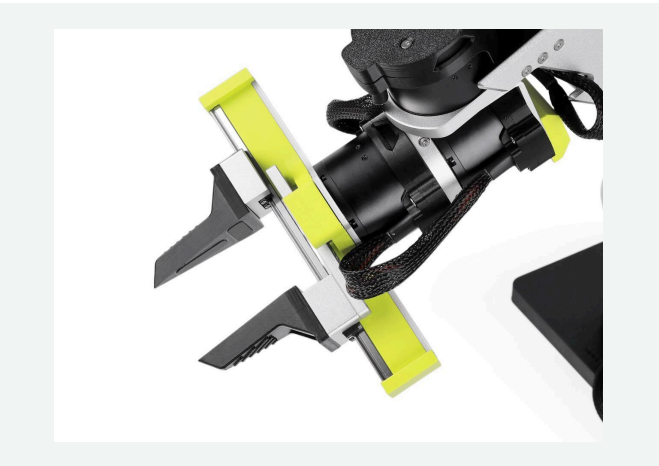
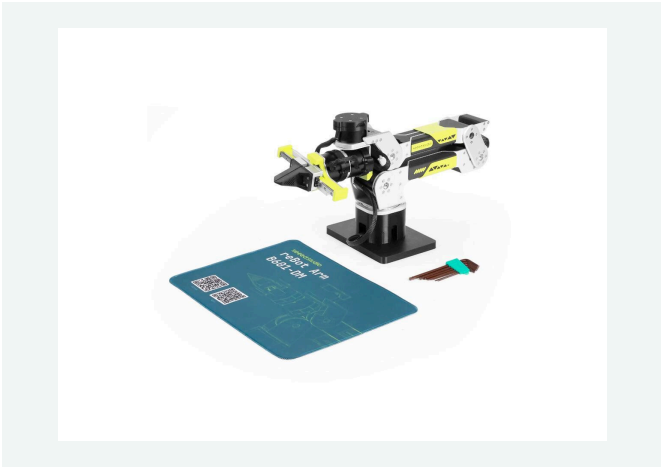
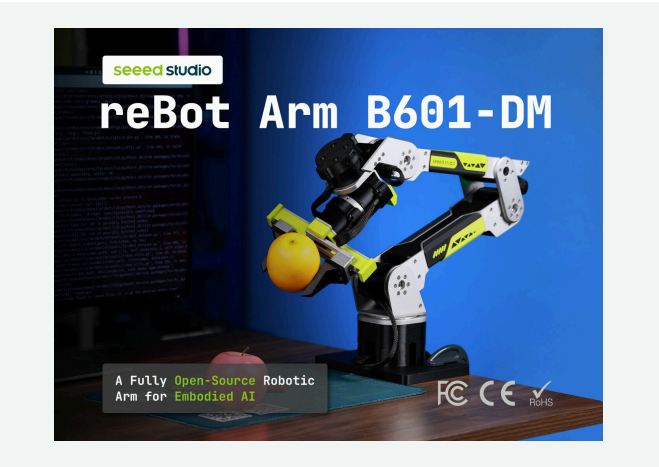
Key Features

- **Plug-and-play bundle:** Assembled robot arm kit plus NVIDIA® Jetson AGX Thor™ Developer Kit, allowing on-site VLA model deployment such as GR00T, PI, OpenVLA etc.
- **Fully Open-Source:** A truly 100% open-source 6+1 DoF robotic arm in terms of hardware and software. It includes BOM lists, hardware drawings, software, and algorithms—everything you need.
- **High-performance motors:** 4 Damiao 4310 motors + 3 Damiao4340P high-torque, low-noise motors ensure stable and precise power for the reBot Arm B601-DM.
- **Framework Compatibility:** Supports mainstream robotics development frameworks such as ROS1/2, Isaac Sim, and LeRobot.
- **Constant Content Updates:** As the project progresses, we will gradually update the latest algorithms and launch a series of completely free courses.

⚠ Buying Guide

This is a **fully assembled version**, ready to use out of the box. If you'd like to **explore teleoperation**, please purchase the "Star Arm 102 (Leader Arm)" and "12V 2A AC/DC Adapter (5-in-1)" together. **A mounting G-clamp is required** to securely fix the robotic arm to a desk/workbench. Please make sure to include one if you do not already have it. All other accessories are optional and can be purchased as needed.

Product Gallery



Specification

	reBot Arm B601-DM Assembled Kit with Gripper
Degrees of Freedom	6+1

Reach	767mm 607mm(without gripper)
Payload	1.5kg (without gripper, 70% reach recommended)
Joint Range of Motion	J1:±150° J2:-220 - 0° J3:-220 - 0° J4:±90° J5:±90° J6:±90° Gripper:-325° - 0°
Repeatability	0.2 mm
Weight	about 4.7kg
Servos	DM-J4340P-2EC x3 DM-J4310-2EC x4
Operating Temperature	-20°~50°
Power	DC 24V 10A
ROS1	✓
Moveit1	✓
ROS2	✓
Moveit2	✓
Python	✓
Lerobot	✓
Isaac Sim	✓
Pinocchio	✓
Open Source	✓

Hardware Overview

Application Devices	Sim-to-Real VLA DLI Course	GraspNet	Voice-control
reBot B601-DM (plus power supply)		✓	✓
reBot B601-RS (plus power supply)	✓		
star 102 Arm (plus power supply)	✓		
NVIDIA Jetson AGX Thor Developer Kit	✓		✓
NVIDIA Jetson Orin Super Developer Kit		✓	
2D Camera (Could use any ET-S231 USB camera from vision accessories list)	✓		
3D Camera (Could use any Orbbec/Re alsense camera from vision accessories list)		✓	✓
reSpeaker			✓

Applications

- **Voice control:** By integrating the reBot arm with the NVIDIA® Jetson AGX Thor™ Developer Kit, you can deploy Large Language Models (LLMs) directly at the edge—enabling real-time, voice-controlled robotic operation (https://wiki.seeedstudio.com/voice_control_rebot_arm). Powered by up to 2070 TFLOPS of AI performance and advanced language understanding capabilities, Jetson AGX Thor interprets natural language commands with high accuracy and seamlessly translates them into precise, real-time control signals for the reBot arm.
- **GraspNet Visual Grasping on Jetson:** Powered by the NVIDIA Jetson AI edge computing platform and reBot Arm B601-DM, this embodied AI vision-based grasping solution (https://wiki.seeedstudio.com/rebot_arm_b601_dm_graspnet_visual_grasping/) enables intelligent robotic interaction with the physical world. By integrating RGB-D depth cameras (Orbbec Gemini 2, <https://www.seeedstudio.com/Orbbec-Gemini-2-3D-Camera-p-6464.html>), it delivers accurate and autonomous 3D perception and robotic grasping. The system integrates advanced technologies including YOLO-based object detection, 3D point cloud processing, the GraspNet-Baseline grasping algorithm, and hand-eye calibration, making embodied AI development more accessible for researchers and developers.
- **Teleoperation & Remote Robotics Control:** The reBot Arm B601 enables intuitive teleoperation for remote manipulation, data collection, and human-in-the-loop training. Combined with AI robotics frameworks like Hugging Face LeRobot, it supports datasets and pre-trained model and policies, and imitation learning workflows that translate human actions into robotic behaviors, accelerating embodied AI development.
- **Assistive Automation for Home Daily Tasks:** The reBot Arm B601 supports the development of assistive robotics for everyday tasks, from pick-and-place to simple household automation. By combining vision and learning-based control, the robot can adapt to unstructured environments and variable objects, helping bring embodied AI into practical, human-centered scenarios.
- **Precision Tasks in Harsh or Constrained Environments:** With ± 0.2 mm repeatability and stable actuation, the B601 is suited for precise tasks in challenging or unsafe environments. Developers can prototype applications such as inspection, sample handling, or fine manipulation under constrained conditions. Its support for dynamic motion execution enables more reliable interaction with uncertain or sensitive objects. This makes it a strong platform for validating robotics solutions before scaling to industrial systems.

Part List

Name	Quantity
Assembled Robotic Arm (with gripper)	1
Hex key wrench set	1
Mouse pad	1

Resource Links

- <https://www.seeedstudio.com/reBot-Arm-B601-DM-Bundle-with-Jetson-Thor.html>
- [reBot Arm Github](#)
- [Tool](#)
- [NVIDIA® Jetson AGX Thor™ Developer Kit Datasheet](#)

Seeed Technology Co.,Ltd.

9F, Building G3, International E City,
Zhongshanyuan Road, Nanshan, Shenzhen, China

Tel: +86 0755-80695676

Web: www.seeed.cc

Shop: www.seeedstudio.com